

MACHINE LEARNING FUNDAMENTALS SERIES

Precision vs. Sensitivity (Recall)

Evaluating Model Reliability and Detection Coverage

Source: Machine Learning Evaluation Metrics Technical Note

1 Overview of Classification Performance Metrics

When evaluating binary classification models, accuracy alone can be misleading, especially under class imbalance or unequal costs of classification errors. **Precision** (Positive Predictive Value) and **Sensitivity** (Recall) isolate two distinct dimensions of model quality:

- **Precision:** Measures the *reliability* of positive predictions.
- **Sensitivity:** Measures the *coverage* or ability to catch all actual positive instances.

2 Precision (Positive Predictive Value)

Formula & Core Question

$$\text{Precision} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Positives (FP)}}$$

Core Question: “Out of all the people predicted by the model to have the disease, what percentage actually have it?”

2.1 Key Characteristics

- **Nature & Focus:** Precision evaluates the **reliability** and trustworthiness of the model’s positive classifications.
- **Denominator Meaning:** The denominator (TP + FP) represents the **entire group of individuals flagged as positive** by the model.
- **High Precision Interpretation:** A high Precision score (e.g., 95%) signifies that out of every 100 individuals flagged as positive, 95 actually have the condition, and only 5 are falsely diagnosed (*false alarms* or False Positives).

2.2 When is High Precision Required?

High Precision is critical in scenarios where the **cost of a false alarm (False Positive)** is **extremely high**.

- **Primary Example — Spam Email Filtering:** Important work or personal emails (which are actually Negative/Legitimate) must not be mistakenly classified as Spam (Positive) and routed to the trash or spam folder.
- **Other Scenarios:** Legal decision support, automated trading execution, high-risk medical intervention recommendation.

3 Sensitivity (Recall)

Formula & Core Question

$$\text{Sensitivity} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$$

Core Question: *“Out of all the people who actually have the disease in real life, what percentage did the model successfully detect?”*

3.1 Key Characteristics

- **Nature & Focus:** Sensitivity evaluates the model’s **coverage** and ability to sweep/capture all positive cases without missing them.
- **Denominator Meaning:** The denominator (TP + FN) represents the **total population of individuals who actually have the disease**.
- **High Sensitivity Interpretation:** A high Sensitivity score (e.g., 98%) means that out of 100 people who truly have the condition, the model successfully identifies 98, leaving only 2 cases missed (*slips through the cracks* or False Negatives).

3.2 When is High Sensitivity Required?

High Sensitivity is essential when the **cost of missing a positive case (False Negative) is catastrophic**.

- **Primary Example — Cancer Diagnosis & Airport Security:** Missing a malignant tumor or an armed threat (False Negative) carries severe, life-threatening risks. Re-screening or further testing a healthy person (False Positive) is far preferable to missing a true threat.

4 Comparative Summary: Precision vs. Sensitivity

The table below summarizes the key differences between Precision and Sensitivity:

Dimension	Precision (Positive Predictive Value)	Sensitivity (Recall)
Formula	$\frac{TP}{TP+FP}$	$\frac{TP}{TP+FN}$
Core Focus	Reliability of positive predictions	Complete detection/coverage of actual positives
Denominator	All predicted positives (TP+FP)	All actual positives (TP + FN)
Key Risk	False Positives (False alarms)	False Negatives (Missed cases)
High-Stakes Domain	Spam filtering, transactional security	Cancer screening, threat detection